

Computer Forensics: Basics

Lecture 1 The Context of Computer Forensics

Adapted from a lecture
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Purdue University 2004

Debate

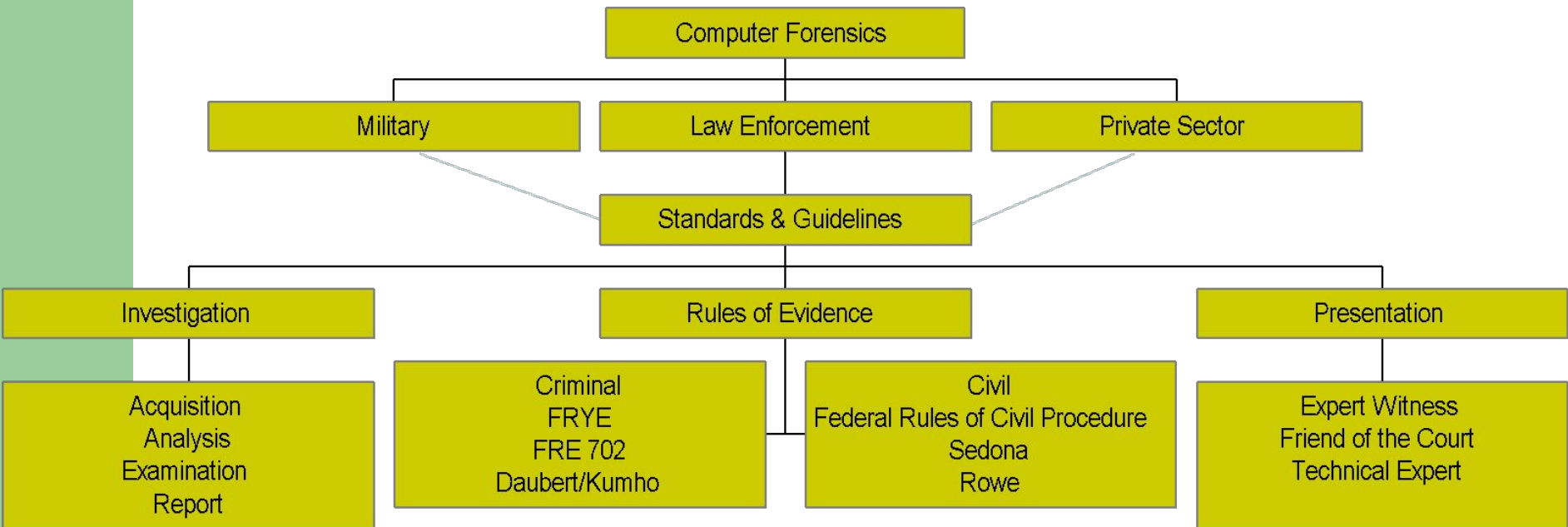
- Is digital forensics a “real” scientific discipline?
 - What is digital forensics
 - How do you define a scientific discipline?
 - Does it really matter?

Learning Objectives

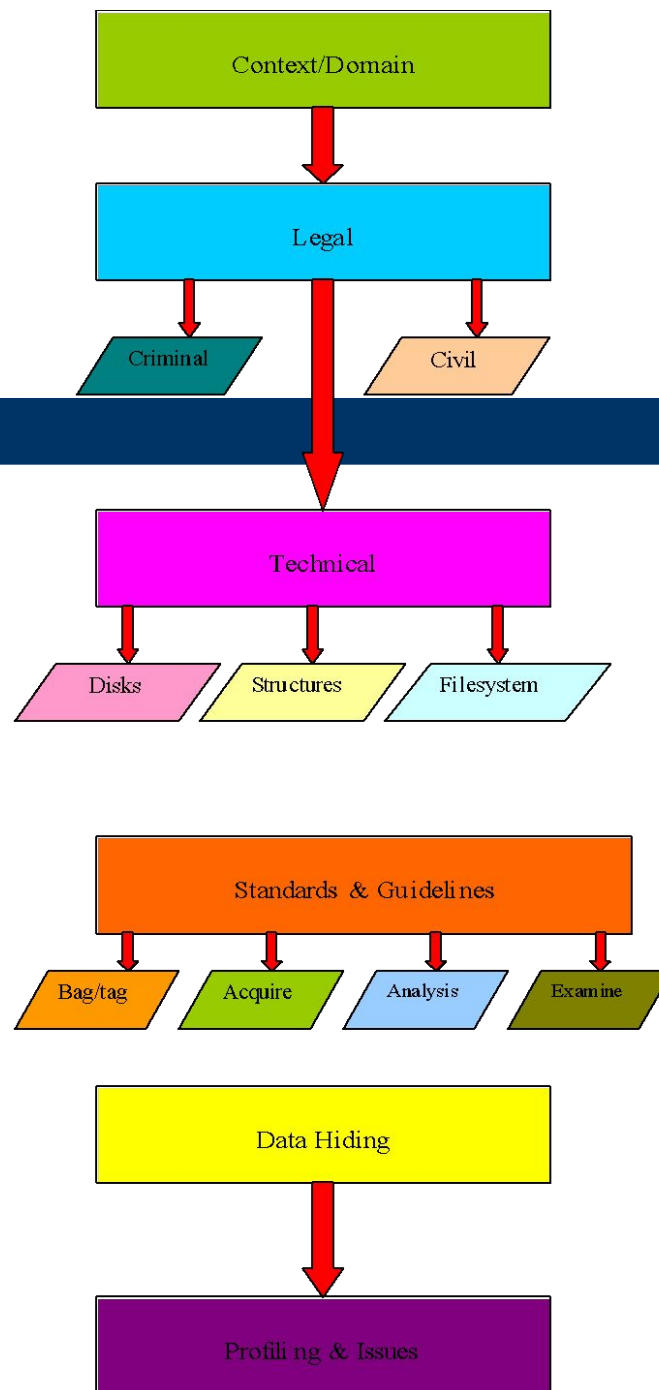
- At the end of this section you will be able to:
 - Describe the science of digital forensics.
 - Categorize the different communities and areas within digital forensics.
 - Explain where computer forensics fits into DFS
 - Describe criminalistics as it relates to the investigative process
 - Discuss the 3 A's of the computer forensics methodology
 - Critically analyze the emerging area of cyber-criminalistics
 - Explain the holistic approach to cyber-forensics

Computer Forensics

Fundamentals



Concept Map



Criminalistics

Criminalistics

- Fancy term for Forensic Science
- Forensic Science
 - The application of science to those criminal and civil laws that are enforced by police agencies in a criminal justice system (Saferstein, 2004)
- Think Sherlock Holmes!!

History & Development

- Francis Galton (1822-1911)
 - First definitive study of fingerprints
- Sir Arthur Conan Doyle (1887)
 - Sherlock Holmes mysteries
- Leone Lattes (1887-1954)
 - Discovered blood groupings (A,B,AB, & O)
- Calvin Goddard (1891-1955)
 - Firearms and bullet comparison
- Albert Osborn (1858-1946)
 - Developed principles of document examination
- Hans Gross (1847-1915)
 - First treatise on using scientific disciplines in criminal investigations.

History & Development

- Edmond Locard (1877-1966)
 - **Principle of Exchange**
 - “..when a person commits a crime something is always left at the scene of the crime that was not present when the person arrived.”
 - The purpose of an investigation is to locate identify and preserve evidence-data on which a judgment or conclusion can be based.
- FBI (1932)
 - National Lab to provide forensic services to all law enforcement agencies in the country

Crime Lab

- Basic services provided
 - Physical Science Unit
 - Chemistry, physics, geology
 - Biology Unit
 - DNA, blood, hair & fiber, body fluids, botanical
 - Firearms Unit
 - Document Examination
 - Photography Unit

Crime Lab

- Optional Services
 - Toxicology Unit
 - Latent Fingerprint Unit
 - Polygraph Unit
 - Voice Print Analysis Unit
 - Evidence Collection Unit (Rather new)

Other Forensic Science Services

- Forensic Pathology
 - Sudden unnatural or violent deaths
- Forensic Anthropology
 - Identification of human skeletal remains
- Forensic Entomology
 - Insects
- Forensic Psychiatry
- Forensic Psychology
- Forensic Odontology
 - Dental
- Forensic Engineering
- *****Digital Forensics*****

Digital Forensic Science

- Digital Forensic Science (DFS):

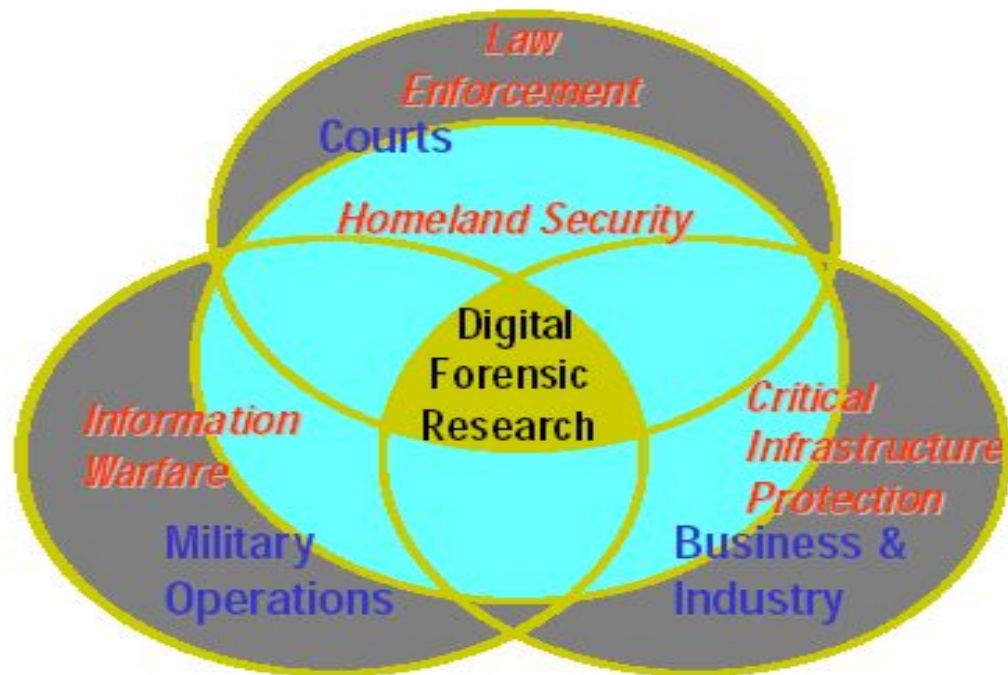
“The use of scientifically derived and proven methods toward the preservation, collection, validation, identification, analysis, interpretation, documentation and presentation of digital evidence derived from digital sources for the purpose of facilitating or furthering the reconstruction of events found to be criminal, or helping to anticipate unauthorized actions shown to be disruptive to planned operations.”

Source: (2001). Digital Forensic Research Workshop (DFRWS)

Communities

- There at least 3 distinct communities within Digital Forensics
 - Law Enforcement
 - Military
 - Business & Industry
 - Possibly a 4th – Academia

Digital Forensic Science



Community Objectives

Table 1 - Suitability Guidelines for Digital Forensic Research

Area	Primary Objective	Secondary Objective	Environment
Law Enforcement	Prosecution		After the fact
Military IW Operations	Continuity of Operations	Prosecution	Real Time
Business & Industry	Availability of Service	Prosecution	Real Time

The Process

- The primary activities of DFS are investigative in nature.
- The investigative process encompasses
 - Identification
 - Preservation
 - Collection
 - Examination
 - Analysis
 - Presentation
 - Decision

Table 2 - Investigative Process for Digital Forensic Science

Identification	Preservation	Collection	Examination	Analysis	Presentation	Decision
Event/Crime Detection	Case Management	Preservation	Preservation	Preservation	Documentation	
Resolve Signature	Imaging Technologies	Approved Methods	Traceability	Traceability	Expert Testimony	
Profile Detection	Chain of Custody	Approved Software	Validation Techniques	Statistical	Clarification	
Anomalous Detection	Time Synch.	Approved Hardware	Filtering Techniques	Protocols	Mission Impact Statement	
Complaints		Legal Authority	Pattern Matching	Data Mining	Recommended Countermeasure	
System Monitoring		Lossless Compression	Hidden Data Discovery	Timeline	Statistical Interpretation	
Audit Analysis		Sampling	Hidden Data Extraction	Link		
Etc.		Data Reduction		Spacial		
		Recovery Techniques				

Subcategories of DFS

- There is a consensus that there are at least 3 distinct types of DFS analysis
 - Media Analysis
 - Examining physical media for evidence
 - Code Analysis
 - Review of software for malicious signatures
 - Network Analysis
 - Scrutinize network traffic and logs to identify and locate

Media Analysis

- May often be referred to as computer forensics.
- More accurate to call it media analysis as the focus is on the various storage medium (e.g., hard drives, RAM, flash memory, PDAs, diskettes etc.)
- Excludes network analysis.

Computer Forensics

- Computer forensics is the scientific examination and analysis of data held on, or retrieved from, computer storage media in such a way that the information can be used as evidence in a court of law.

Computer Forensic Activities

- Computer forensics activities commonly include:
 - the **secure** collection of computer data
 - the **identification** of suspect data
 - the **examination** of suspect data to determine details such as origin and content
 - the **presentation** of computer-based information to courts of law
 - the **application** of a country's laws to computer practice.

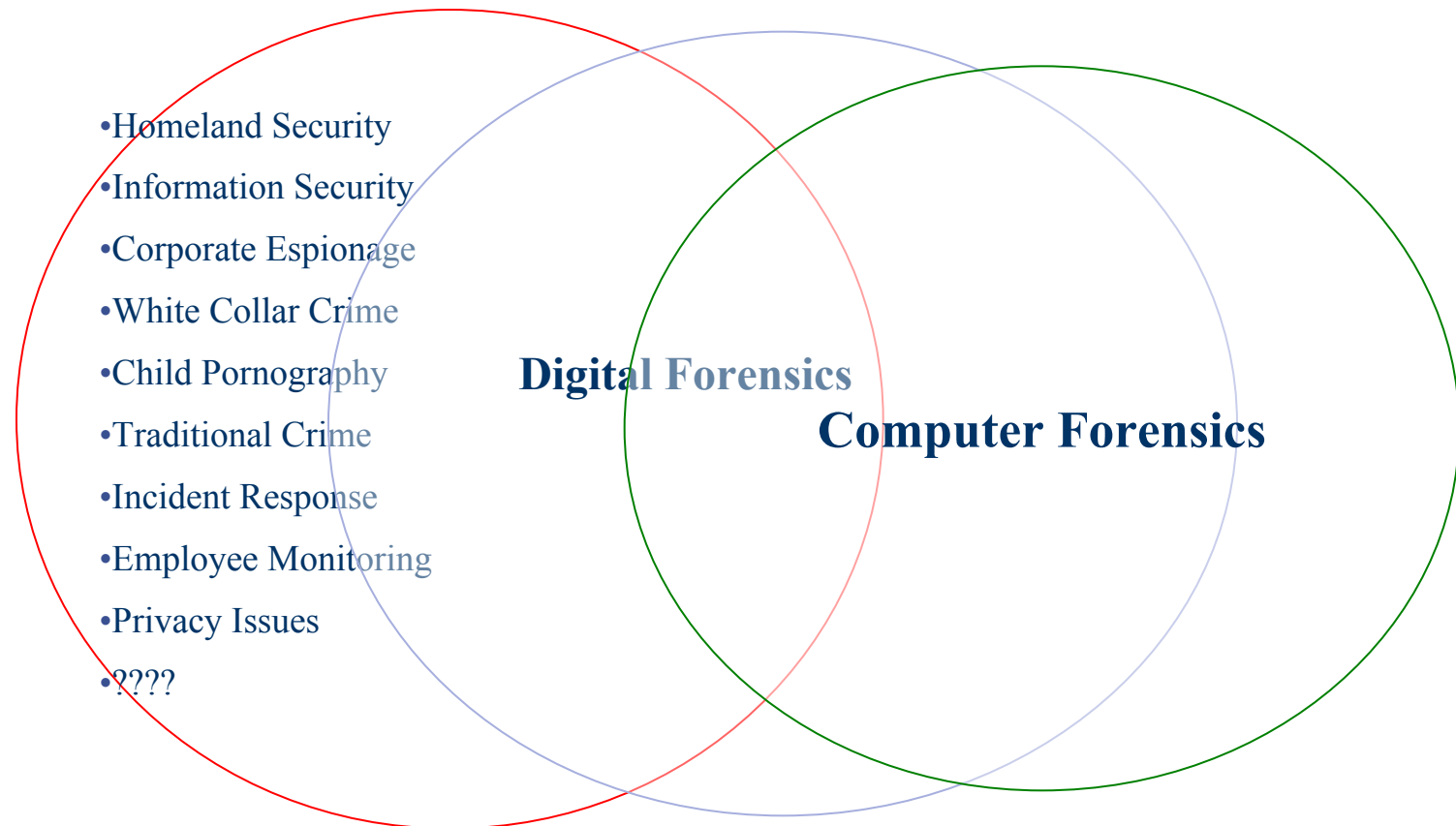
The 3 As

- The basic methodology consists of the 3 As:
 - ***Acquire*** the evidence without altering or damaging the original
 - ***Authenticate*** the image
 - ***Analyze*** the data without modifying it

Computer Forensics - History

- 1984 FBI Computer Analysis and Response Team (CART)
- 1991 International Law Enforcement meeting to discuss computer forensics & the need for standardized approach
- 1997 Scientific Working Group on Digital Evidence (SWGDE) established to develop standards
- 2001 Digital Forensic Research Workshop (DFRWS) development of research roadmap
- 2003 Still no standards developed or corpus of knowledge (CK)

Context of Computer Forensics

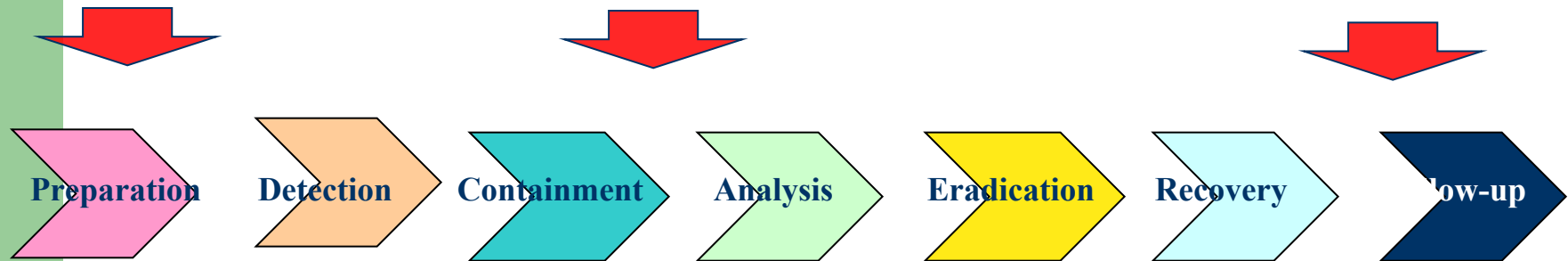


Fit with Information Assurance

- Computer Forensics is part of the incident response (IR) capability
- Forensic “friendly” procedures & processes
- Proper evidence management and handling
- IR is an integral part of IA

Incident Response Methodology (PDCAERF)

Digital Forensics/Evidence Management



Feed Back

(PDCAERF)

- Preparation
 - Being ready to respond
 - Procedures & policies
 - Resources & CSIRT creation
 - Current vulnerabilities & counter-measures
- Detection/Notification
 - Determining if an incident or attempt has been made
 - IDS
 - Initial actions/reactions
 - Determining the scope
 - Reporting process

(PDCAERF)

- Containment
 - Limit the extent of an attack
 - Mitigate the potential damage & loss
 - Containment strategies
- Analysis & Tracking
 - How the incident occurred
 - More in-depth analysis of the event
 - Tracing the incident back to its source

(PDCAERF)

- Eradication/ Repair-Recovery
 - Recovering systems
 - Getting rid of the causes of the incident, vulnerabilities or the residue (rootkits, trojan horses etc.)
 - Hardening systems
 - Dealing with patches

(PDCAERF)

- Follow-up
 - Review the incident and how it was handled
 - Postmortem analysis
 - Lessons learned
 - Follow-up reporting

Challenges

- Eric Holder, *Deputy Attorney General of the United States Subcommittee on Crime of the House Committee on the Judiciary and the Subcommittee on Criminal Oversight of the Senate Committee on the Judiciary*:
 - *Technical challenges* that hinder law enforcement's ability to find and prosecute criminals operating online;
 - *Legal challenges* resulting from laws and legal tools needed to investigate cybercrime lagging behind technological, structural, social changes; and
 - *Resource challenges* to ensure we have satisfied critical investigative and prosecutorial needs at all levels of government.

Challenges

- NIJ 2001 Study
 - There is near-term window of opportunity for law enforcement to gain a foothold in containing electronic crimes.
 - Most State and local law enforcement agencies report that they lack adequate training, equipment and staff to meet their present and future needs to combat electronic crime.
 - Greater awareness of electronic crime should be promoted for all stakeholders, including prosecutors, judges, academia, industry, and the general public.

General Challenges

- Computer forensics is in its infancy
- Different from other forensic sciences as the media that is examined and the tools/techniques for the examiner are products of a market-driven private sector
- No real basic theoretical background upon which to conduct empirical hypothesis testing
- No true professional designations
- Proper training
- At least 3 different “communities” with different demands
- Still more of a “folk art” than a true science

Legal Challenges

- Status as scientific evidence??
- Criteria for admissibility of novel scientific evidence (*Daubert v. Merrell*)
 - Whether the theory or technique has been reliably tested;
 - Whether the theory or technique has been subject to peer review and publication;
 - What is the known or potential rate of error of the method used; and
 - Whether the theory or method has been generally accepted by the scientific community.
- Kumho Tire extended the criteria to technical knowledge

Specific Challenges

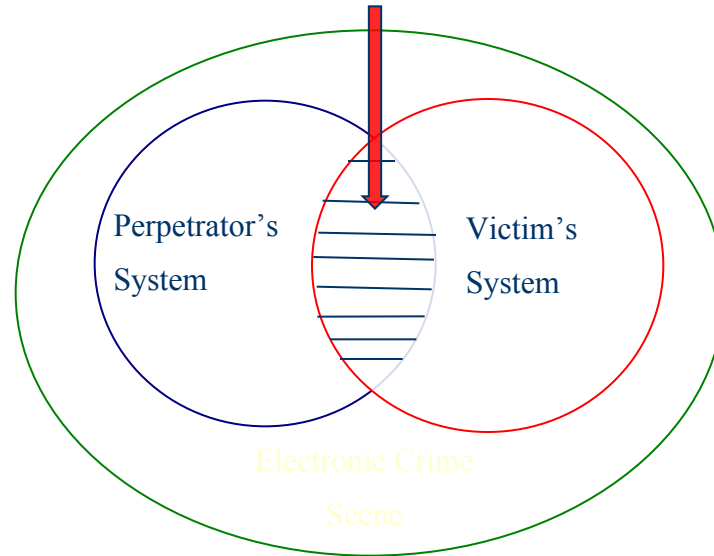
- No International Definitions of Computer Crime
- No International agreements on extraditions
- Multitude of OS platforms and filesystems
- Incredibly large storage capacity
 - 100 Gig Plus
 - Terabytes
 - SANs

Specific Challenges

- Small footprint storage devices
 - Compact flash
 - Memory sticks
 - Thumb drives
 - Secure digital
- Networked environments
- RAID systems
- Grid computing
- Embedded processors
- Other??

Specific Challenges

- Where is the “crime scene?”



Specific Challenges

- What constitutes evidence??
- What are we looking for??

Summary

- DFS is a sub-discipline of criminalistics
- DFS is a relatively new science
- 3 Communities
 - Legal, Military, Private Sector/Academic
- DFS is primarily investigative in nature
- DFS is made up of
 - Media Analysis
 - Code Analysis
 - Network Analysis

Summary

- Computer Forensics is a sub-discipline within DFS
- Computer Forensics is part of an IR capability
- 3 A's of the Computer Forensic Methodology
- There are many general and specific challenges
- There is a lack of basic research in this area
- Both DFS and Computer Forensics are immature emerging areas